

CHECKING THE WHOLE EDIFICE: AN INDEPENDENT LANGEVIN SIMULATION OF $\dot{T}$, $\dot{X}^2$, $\dot{Y}^2$

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ABSTRACT. Every result in [2, 3] — the exact $Tv^2/\mu = \frac{1}{4}$ heating/cooling boundary, the Maxwellian-corrected diffusion tensor, the $\langle \dot{R}_-^2 \rangle \geq 0$ finding, and the counterintuitive sign reversal of $\langle \dot{X}^2 \rangle$ away from the turning points — was derived through symbolic Itô calculus and checked only against internal algebraic identities (linearity of Itô’s lemma, reduction to known limits). None of it had been checked against a direct simulation of the underlying physical process. This note does that: we simulate the full anisotropic Langevin equation for a test particle’s velocity directly in Cartesian coordinates, with drift and diffusion set by [1]’s Maxwellian-averaged coefficients, and measure the resulting secular drift of the osculating elements T, X^2, Y^2 by ensemble averaging. Doing this exposed a genuine implementation bug — a naive planar simulation supplies only one of the two perpendicular noise directions that $\text{Tr } D = D_{\parallel} + 2D_{\perp}$ requires, since a real bath kicks a particle out of its orbital plane as readily as it kicks it sideways within that plane, even though the plane itself is only tracked to leading order — which silently produces the wrong Jensen/Itô correction to $\langle \dot{h} \rangle$ and hence to $\langle \dot{T} \rangle$. With the missing channel restored, four independent tests — $\langle \dot{T} \rangle$ on both sides of the $\frac{1}{4}$ boundary, $\langle \dot{R}_-^2 \rangle$ at an extreme (high-eccentricity, cold-bath) point, and the sign-reversed $\langle \dot{X}^2 \rangle$ at a point selected specifically because the analytic formula predicts it negative — all agree with the analytic predictions to within statistical error (largest deviation 0.4σ). This constitutes the first check of this line of work against anything other than itself.

1. THE SIMULATOR

A test particle’s position and velocity evolve as

$$(1) \quad \dot{r} = v, \quad dv = \left(-\frac{\mu r}{|r|^3} - |A|\hat{v} \right) dt + \sqrt{2D_{\parallel}} dt \eta_{\parallel} \hat{v} + \sqrt{2D_{\perp}} dt \eta_{\perp} \hat{e}_{\perp} + \sqrt{2D_{\perp}} dt \eta_z \hat{z},$$

$\eta_{\parallel}, \eta_{\perp}, \eta_z$ independent standard normals resampled at every step, $D_{\parallel}(v), D_{\perp}(v), |A|(v)$ the Maxwellian-averaged forms of [1] at the instantaneous speed. The particle’s motion is confined to the $z = 0$ plane to leading order (gravity and friction act only on the in-plane components), but the energy $h = \frac{1}{2}(|v|^2 + v_z^2) - \mu/r$ includes the accumulated out-of-plane velocity v_z , while r, ℓ, T, X^2, Y^2 are computed from the in-plane components alone. This is the correct leading-order treatment for measuring the instantaneous secular drift over a time short compared to the timescale on which the orbital plane itself would visibly tilt.

Remark (The bug, and why it is easy to make). An initial implementation supplied only $\hat{v}$ and the single in-plane $\hat{e}_{\perp}$ direction, omitting $\hat{z}$ entirely — a natural simplification, since the orbit itself is exactly 2D and ℓ, T, X^2, Y^2 genuinely need no third dimension. But $h = \frac{1}{2}|v|^2 - \mu/r$ is the kinetic energy of the full 3D velocity vector, and the Maxwellian bath does not know or care that the orbit happens to be planar: it kicks the particle out of the plane with the same $D_{\perp}$ it uses for the in-plane perpendicular kick. Dropping this channel silently

halves the Jensen/Itô correction $\text{Tr } D = D_{\parallel} + 2D_{\perp} \rightarrow D_{\parallel} + D_{\perp}$ appearing in $\langle \dot{h} \rangle$ (Corollary in [3]), which propagates into every downstream quantity. This was caught by comparing $\langle \dot{h} \rangle$'s measured mean directly against the analytic prediction at a point where the sign was unambiguous, isolating the discrepancy to the diffusion (not friction) contribution via a separate deterministic-only (noise-free) control run.

2. VALIDATION 1: $\langle \dot{T} \rangle$ ON BOTH SIDES OF THE $\frac{1}{4}$ BOUNDARY

Using $T = 1, \mu = 1, e = 0.8, r = 1.7$ ($Tv^2/\mu \approx 0.176 < \frac{1}{4}$, so [2]'s Theorem predicts a sign change in χ):

σ	χ	analytic $\langle \dot{T} \rangle$	measured (2M trajectories)
0.10	2.97	-0.00149	-0.00184 ± 0.00195
0.30	0.99	+0.07383	$+0.07329 \pm 0.00379$

Both signs and magnitudes match within statistical error.

3. VALIDATION 2: $\langle \dot{T} \rangle$ DEEP IN THE UNCONDITIONAL-HEATING REGIME

At $r = 1.5$ ($Tv^2/\mu = \frac{1}{3} > \frac{1}{4}$, robustly heating for any bath temperature), $\sigma = 0.3$: analytic $\langle \dot{T} \rangle = 0.03779$, measured 0.03654 ± 0.00419 (0.3σ).

4. VALIDATION 3: $\langle \dot{R}_{-}^2 \rangle \geq 0$ AT AN EXTREME POINT

At $e = 0.95$, exactly at apoapsis, $\sigma = 0.05$ (a case from [3]'s aggressive sweep): analytic $\langle \dot{R}_{-}^2 \rangle = 2.973$, measured 2.970 ± 0.012 (0.03σ).

5. VALIDATION 4: THE COUNTERINTUITIVE SIGN REVERSAL OF $\langle \dot{X}^2 \rangle$

The most important check: at $e = 0.95, \theta = 173$ (in the interior band identified in [3] where $\langle \dot{X}^2 \rangle$ is negative despite $X^2 \geq 0$ trivially and the naive turning-points-only picture suggesting otherwise), analytic $\langle \dot{X}^2 \rangle = -0.0573$, measured -0.0585 ± 0.0067 (0.17σ). The counterintuitive sign is confirmed by direct simulation, not merely by the symbolic derivation that first produced it.

6. SCOPE

- This validates the Itô calculus and the (h, ℓ) SDE machinery of [2, 3] directly, independent of the symbolic derivations — it does not re-validate [1]'s own Rosenbluth-potential physics, which rests on separate (already-cited) grounds.
- All tests use short simulated times (secular drift measured well before the orbit itself evolves appreciably) and equal masses, matching the scope of every note this validates.
- The three open items of [3] (the covariance bound, $\langle \dot{R}_{-}^2 \rangle$'s closed form, Y 's boundary) are untouched by this note; it confirms the numerics feeding into them are trustworthy, not the missing proofs themselves.

AI ASSISTANCE DISCLOSURE

Large language model tools (Anthropic’s Claude) were used in the derivation, simulation, and L^AT_EX preparation of this note. This includes: implementation of the Langevin simulator; diagnosis of an initial, substantial ($\sim$ tens-of-percent to order-of-magnitude, and in one case sign-obscuring) mismatch between simulated and analytic $\langle \dot{T} \rangle$, isolated first by ruling out numerical integration error (via a noise-free control run matching to 10^{-7}), then by ruling out the friction/drift term (via a deterministic-friction-only run matching the analytic $v \cdot A$ contribution essentially exactly), which together isolated the discrepancy to the diffusion channel and led to identifying the missing out-of-plane noise direction; and the four validation runs reported above, each independently seeded and cross-checked against the analytic formulas already established in [2, 3]. All numerical claims were independently verified by the author.

REFERENCES

- [1] J. Ebert, *Doing the Actual Average: Rosenbluth Potentials Close the Residual*, Preprint (2026).
- [2] J. Ebert, *The Cancellation That Wasn’t: Friction, Diffusion, and an Exact Heating/Cooling Boundary for Orbits in a Maxwellian Bath*, Preprint (2026).
- [3] J. Ebert, *The Rest of the Cone: X and Y Under a Genuine Maxwellian Bath, at General Orbital Phase*, Preprint (2026).

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